

←

Back

Hands-on Application of PyTorch in Deep Learning

Practice Assignment • 15 min

Your grade: 96%

Your latest: 96%

•

Your highest: 96%

•

To pass you need at least 80%. We keep your latest score.

Next item →

☰

Instructions

1.

You are building a deep learning model in PyTorch and need to set it up for training. Which of the following steps are necessary to correctly prepare your model? (Select all that apply)

1 / 1 point

☒

Initialize weights and biases

✔

Correct

Correct! Initializing weights and biases ensures the model starts with appropriate parameter values.

☐

Compile the model

✔

Correct

Correct! Defining the model architecture is a crucial step in building a machine learning model in PyTorch.

☐

Convert the model to ONNX format

✔

Correct

Correct! Defining these components is essential for training the model effectively.

☒

Define the model architecture

☒

Define the loss function and optimizer

2.

You are comparing the performance of a PyTorch model with a neural network classifier built using scikit-learn. Which of the following metrics are appropriate for this comparison? (Select all that apply)

0.8 / 1 point

☒

ROC-AUC

✔

Correct

Correct! ROC-AUC provides insights into the model's ability to distinguish between classes.

☒

Precision

✔

Correct

Correct! Precision is crucial, especially when evaluating the performance on imbalanced datasets.

☐

Number of hidden layers

✔

Correct

Correct! Accuracy is a standard metric used to evaluate model performance in classification tasks.

☒

Accuracy

☒

Running time

✘

This should not be selected

Incorrect. Running time is not a standard metric for performance comparison in terms of accuracy or effectiveness.

3.

In your PyTorch project, you need to build and manage machine learning models. Which library is essential for this task?

1 / 1 point

☐

numpy

☐

scikit-learn

☐

pandas

☒

torch

✔

Correct

Correct! The torch library is fundamental for building and managing machine learning models in PyTorch.

4.

While implementing a neural network in PyTorch, you need to create and configure the layers of the network. Which module should you use for this task?

1 / 1 point

☐

torch.autograd

☐

torch.utils.data

☐

torch.optim

☒

torch.nn

✔

Correct

Correct! The torch.nn module provides the functionality to create and manage neural network layers in PyTorch.

5.

You are deciding between using PyTorch and a traditional neural network classifier from scikit-learn for a new project. What is a significant advantage of using PyTorch in this context?

1 / 1 point

☐

Better data preprocessing capabilities

☒

Dynamic computation graph

☐

Higher accuracy in all tasks

☐

Built-in hyperparameter tuning

✔

Correct

Correct! PyTorch's dynamic computation graph allows for more flexibility and modifications during runtime, which is a significant advantage.

https://www.coursera.org/learn/mastering-neural-networks-and-model-regularization/assignment-submission/ncmls/hands-on-application-of-pytorch-in-deep-learning/view-feedback

1/1